

12 Probability distributions

Subject content	Additional information
12.1 Know the use and validity of distributions which could be appropriate in a particular real world situation: binomial, normal, Poisson and exponential.	
12.2 Evaluate the mean and variance of linear combinations of independent random variables through knowledge that if X_i are independently distributed (μ_i, σ_i^2) then $\sum a_i X_i$ is distributed $(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2)$.	Students should know that $E(aX \pm bY) = aE(X) \pm bE(Y)$
12.3 Evaluate probabilities for linear combinations of two or more independent normal distributions and apply this knowledge to practical situations.	